

Finesse3 Optics Simulator

User Manual v1.1.0

Frequency-domain optical interferometer design & simulation

1. Getting Started

Requirements: Windows 10/11, Miniconda, Finesse 3 ($\geq 3.0a33$) in a conda env named finesse.

```
conda create -n finesse python=3.10 -y
conda activate finesse
pip install finesse numpy scipy networkx flask
```

Launch: double-click run.bat. The launcher finds a free port and opens your browser at <http://localhost:5000>.

2. Interface Overview

Panel	Purpose
Library (left)	Drag components onto the canvas
Canvas (center)	Visual layout of the optical setup
Properties (right)	Edit parameters of the selected item
Output (bottom)	KatScript preview, charts, CCD images

Shortcuts: Shift+drag locks movement direction. Delete removes the selected item.

3. Building a Model

1. Drag a component from the left library onto the canvas.
2. Connect ports by dragging from one component's red port dot to another.
3. Repeat to build your optical layout.

Available Components

Icon	Type	Description
L	Laser	Laser source

M	Mirror	Reflective mirror
BS	Beamsplitter	4-port splitter
Ln	Lens	Thin lens
EOM	Modulator	Electro-optic modulator
Iso	Isolator	Optical isolator
PD	Photodetector	Power detector
CCD	CCD	Beam profile camera
G	Gauss	Beam parameter definition
Cav	Cavity	Optical cavity
AD	Amp. Detector	Amplitude detector
?	Custom	Other KatScript component

Manipulating Components

- **Drag** to move.
- **Flip** (Flip H / Flip V buttons in Properties).
- **Rotate** (+-90, +-45, or custom angle in Properties).
- **Right-click** for context menu, or **Delete** key to remove.

4. Editing Components

Click a component. The Properties panel shows:

Name — unique ID, e.g. m1.

Parameters — R, T, f, phi (numbers or {var} expressions).

Advanced — click to show alpha, Rc, etc.

Extra param — add any KatScript parameter.

Blue text = active variable reference. Example: $R=\{bs_r\}$, $T=1-\{bs_r\}$.

5. Editing Connections

Click a connection line. Properties shows:

Port selectors — switch which ports are linked.

Name — space ID.

Length (L) — optical path length in metres.

Extra param — e.g. nr for refractive index.

6. Variables and Expressions

Click Vars in the toolbar to open the Variables panel. Define:

```
Name: arm_len    Value: 3.0
Name: R_bs       Value: 0.5
```

Use {name} in any parameter field: $R=\{R_bs\}$, $L=\{arm_len\}$.

When importing Python scripts, expressions using numpy, math, etc. are evaluated server-side and the results become variables.

7. Running Simulations

Run — static (noaxis). Shows KatScript, PD readouts, CCD images (if Gauss/Cavity is upstream of the CCD).

Sweep — parameter scan (xaxis). Select a parameter, set range and steps. Chart.js plots detector outputs vs. the swept parameter. Demodulated detectors (pd1/pd2) show amplitude and phase.

8. Importing Models

Paste KatScript or an entire Python/Jupyter cell into the import text area and click Import.

Directly from Jupyter

Copy your whole Jupyter cell and paste it. The importer handles everything automatically:

- Strips ``import`, `print()`, `model.run()`, and non-model code.`
- Evaluates Python variables (including ``numpy`` expressions).
- Parses every ``model.parse("..."`)` block into components and connections.
- Preserves ``{var}`` references in parameters.

You can paste code with multiple `model.parse()` calls, debug prints, or any extra lines — it all works.

Example 1 — Basic Michelson (mod1)

Paste these KatScript lines:

```
l L0 P=1
s s0 L0.p1 bs1.p1 L=0.5
bs bs1 R=0.5 T=0.5
s s1 bs1.p2 m1.p1 L=3
m m1 R=0.99 T=0.01
s s2 bs1.p3 m2.p1 L=3.001
m m2 R=0.99 T=0.01
s s3 bs1.p4 pd1.p1 L=0.5
pd pd1
```

Light splits at bs1, travels to m1 (3 m) and m2 (3.001 m), recombines, interference measured at pd1.

Example 2 — With Variables (mod2)

Paste the whole cell:

```
arm_len = 3.0
delta_len = 0.001
bs_r = 0.5
m_r = 0.99
model.parse("""
l L0 P=1
s s0 L0.p1 bs1.p1 L=0.5
bs bs1 R={bs_r} T=1-{bs_r}
s s1 bs1.p2 m1.p1 L={arm_len}
m m1 R={m_r} T=1-{m_r}
s s2 bs1.p3 m2.p1 L={arm_len+delta_len}
m m2 R={m_r} T=1-{m_r}
s s3 bs1.p4 pd1.p1 L=0.5
pd pd1
""")
```

Vars panel shows `arm_len`, `delta_len`, `bs_r`, `m_r`. Edit any to tune the whole model.

Example 3 — Computed Variables (mod3)

Arm length computed from wavelength with numpy:

```
import numpy as np
wavelength = 1064e-9
arm_len = int(3.0 / wavelength) * wavelength
bs_angle = np.rad2deg(np.arcsin(0.1))
m_refl = 0.99
model.parse("""
l L0 P=1
s s0 L0.p1 bs1.p1 L=0.5
bs bs1 R={m_refl} T=1-{m_refl} alpha={bs_angle}
s s1 bs1.p2 m1.p1 L={arm_len}
m m1 R={m_refl} T=1-{m_refl}
s s2 bs1.p3 m2.p1 L={arm_len}
m m2 R={m_refl} T=1-{m_refl} phi=90
s s3 bs1.p4 pd1.p1 L=0.5
pd pd1
""")
```

After import: wavelength = 1.064e-07, arm_len (integer wavelengths near 3 m), bs_angle (from np.rad2deg), m_refl = 0.99. Try sweeping m2.phi 0–360 degrees.

9. Exporting Models

Export .kat — plain-text KatScript file. Header lists any auto-renamed spaces.

Export .ipynb — ready-to-run Jupyter notebook.

10. Custom Components

Finesse has many component types beyond the built-in list (dbs, grating, squeezer). Use the Custom component:

1. Drag Custom (grey "?" icon) onto the canvas and enter the number of ports.
2. In Properties, set Kat Type (e.g. dbs) and Name.
3. Use Extra param to add any KatScript parameters.

11. Saving and Loading

Click Save, enter a name. The model (all component positions, angles, connections, variables, and parameters) is saved to the examples/ folder as a JSON file.

Click Example to browse saved models. Click any entry to load it, or the x button to delete.

12. CCD Imaging

1. Add a CCD connected to an optical node.
2. Add a Gauss defining beam parameters (w0, z) upstream of the CCD.
3. Click Run, then View CCD in the output panel.

The beam profile image opens in a new window.

13. Jupyter Integration

Click Jupyter in the output panel to open a built-in code editor. The current model is available as the model variable:

```
print(model.m1.R)
sol = model.run("noxaxis()")
print(sol['pd1'])
```

14. Light / Dark Theme

Click the sun/moon icon in the header to toggle the theme.